

1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

```
C:\Users\margi>ping www.youtube.com

Pinging www.youtube.com [2001:4860:482c:400::] with 32 bytes of data:
Reply from 2001:4860:482c:400::: time=77ms
Reply from 2001:4860:482c:400::: time=56ms
Reply from 2001:4860:482c:400::: time=58ms
Reply from 2001:4860:482c:400::: time=48ms

Ping statistics for 2001:4860:482c:400:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 48ms, Maximum = 77ms, Average = 59ms
```

2.Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Hint: Each hop represents a network device your data passes through to reach the destination.

```
C:\Users\margi>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [23.218.249.114]
over a maximum of 30 hops:

  1    7 ms    4 ms    3 ms  Unit [192.168.1.1]
  2   32 ms   30 ms   27 ms  10.240.8.38
  3    *      *      *      Request timed out.
  4   37 ms   55 ms   39 ms  182.79.249.53
  5   86 ms   63 ms   73 ms  116.119.35.80
  6   60 ms   52 ms   48 ms  ae7.r02.bom01.iem.netarch.akamai.com [23.56.136.206]
  7   90 ms   57 ms   65 ms  ae11.tata-bom3.netarch.akamai.com [104.70.118.135]
  8   50 ms   54 ms   52 ms  a23-218-249-114.deploy.static.akamaitechnologies.com [23.218.249.114]

Trace complete.
```

3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.

```
C:\Users\margi>nslookup www.zomato.com
Server: UnKnown
Address: fe80::26de:8aff:fe79:8d11

Non-authoritative answer:
Name: e70929.dsca.akamaiedge.net
Addresses: 2600:140f:1e00::1737:f4c3
           2600:140f:1e00::1737:f4d0
           23.218.249.129
           23.218.249.115
Aliases: www.zomato.com
          zomato.edgekey.net
```

4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

```
C:\Users\margi>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

5. Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: Give a specific example for each tool based on your own test results above.

- **Ping** is useful for checking if a website or device is reachable. For example, my ping test to **YouTube** received replies, showing that the connection was working.
- **Traceroute** is useful for finding where a connection is slow or failing. My **tracert test to Flipkart** showed the different hops between my computer and the website.
- **Nslookup** is useful for checking **DNS problems**. My test for **Zomato** returned its IP address, showing that DNS was working.
- For a **website not loading**, I would first use **ping** to check connectivity.
- For a **slow connection**, I would use **traceroute** to find problematic hops.
- If I suspect a **DNS issue**, I would use **nslookup** to check whether the website name is resolving to an IP address.